# Ethan Shen

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## Education

**University of Washington** Seattle, WA
PhD in Computer Science and Engineering Sep. 2025 - Jun. 2029
**Advisor**: Ali Farhadi

**University of Washington** Seattle, WA
B.S. in Computer Science, Minor in Mathematics (GPA: 3.96/4.00) Aug. 2022 - Jun. 2025

## Publications

*Perception Tokens Enhance Visual Reasoning in Multimodal Language Models*
Mahtab Bigverdi, Zelun Luo, Cheng-Yu Hsieh, **Ethan Shen**, Dongping Chen, Linda Shapiro, Ranjay Krishna.
CVPR 2025.

*Superposed Decoding: Multiple Generations from a Single Autoregressive Inference Pass.*
**Ethan Shen**, Alan Fan, Sarah M Pratt, Jae Sung Park, Matt Wallingford, Sham M Kakade, Ari Holtzman, Ranjay Krishna, Ali Farhadi, Aditya Kusupati.
NeurIPS 2024.

*Are "Hierarchical" Visual Representations Hierarchical?*
**Ethan Shen**, Ali Farhadi, Aditya Kusupati.
Workshop on Symmetry and Geometry in Neural Representations @ NeurIPS 2023.

## Research Experience

**Allen Institute of Artificial Intelligence** Seattle, WA
Research Intern (Advisor: Prof. Tim Dettmers) Feb. 2025 - Present
- Leading development of an open-source alternative to closed-source coding agents such as Claude Code.
- Experimented with network stacking for low-cost LLM scaling and implemented distributed training for 100M to 7B parameter models on a trillion-token scale.

**Reasoning, AI, Vision Lab @ UW** Seattle, WA
Research Assistant (Advisors: Prof. Ali Farhadi, Prof. Ranjay Krishna) Jun. 2023 - Present
- Pioneered latent token representations for multimodal reasoning in vision-language models like Qwen and LLaVA.
- Created LLM decoding algorithm to improve inference efficiency and accuracy for multi-draft generation.
- Developed suite of hierarchical vision datasets for understanding complex visual hierarchies.

**Sensor Systems Lab @ UW** Seattle, WA
Research Assistant (Advisor: Prof. Joshua Smith) Jun. 2022 - Jun. 2023
- Designed and built an acoustic levitator, a tool that uses ultrasonic sound to levitate fragile objects for scientific experiments.
- Programmed mathematical algorithms in Python and C++ to simulate, predict, and control the rotation of objects in the levitator, with a 15x speedup compared to existing simulations.

## Work Experience

**Amazon** Seattle, WA
Software Engineer Intern Jun. 2024 - Sep. 2024
- Built a scalable GenAI pipeline with AWS Bedrock, Lambda, and DynamoDB to automatically process up to 15,000 financial documents a month, saving over 1200 hours of manual work per year.
- Implemented fine tuning and heurestic filtering to achieve 96% accuracy on an internal invoice dataset.

**Papyrus** San Francisco, CA
Machine Learning Engineer Intern Dec. 2023 - Mar. 2024
- Worked at pre-seed startup and created a first-of-its-kind feature to label speakers in transcripts using long-context LLMs.
- Developed ASR pipeline for transcription/translation with Whisper and improved diarization with audio processing tools like Demucs and Pyannote.

## Professional Services

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| 2025 | **NeurIPS Reviewer (2025)** |
| 2024 | **EMNLP Reviewer (2024, 2025)** |
| 2024 | **UW CSE: Student Interviewer for Faculty Hiring** |
| 2023 | **UW CSE: Teaching Assistant** |
| 2022-2023 | **UW CSE: Student Advisory Council Officer** |

## Presentations

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| Oct. 2024 | **UW RAIVN Lab (Host: Ali Farhadi)**, Topic: Superposed Decoding |
| Aug. 2024 | **AWS AI Labs CodeGen (Host: Zijian Wang)**, Topic: Superposed Decoding |
| Aug. 2024 | **Amazon LLM Reasoning Group (Host: Linbo Liu)**, Topic: Superposed Decoding |
| Jul. 2024 | **AWS AI Labs Quicksight (Host: Patrick Ng)**, Topic: Superposed Decoding |

## Honors & Awards

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| 2023 | **NeurIPS Travel Award**, Conference for Neural Information Processing Systems |
| 2022 | **FEEA Merit Scholarship**, US Department of Veterans Affairs |
| 2021 | **42/45 Score (Top 8% Internationally)**, IB Diploma Programme |
| 2021 | **National Merit Finalist (Top 1% Nationally)**, National Merit Program |